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Cross-Slice Transformer with Adaptive Slice Fusion for Lymph Node Segmentation in CT Imaging

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ABSTRACT Accurate evaluation of lymph nodes on computed tomography (CT) scans is essential for cancer staging and treatment planning. However, automating this task remains challenging because many nodes are very small, contrast with surrounding tissue is frequently low, and malignant examples constitute only a minuscule fraction of all voxels—an extreme class-imbalance problem. We propose a Cross-Slice Transformer (CST) that learns adaptive slice importance through statistical descriptor extraction and transformer-based attention, combined with consistency regularization for improved inter-slice modeling in 2.5D CT analysis. CST employs adaptive slice fusion that automatically weights three consecutive CT slices based on their information content, enabling effective volumetric context integration while maintaining computational efficiency. The CST module is integrated with a U-Net decoder for end-to-end segmentation training. We evaluate CST against four state-of-the-art segmentation architectures (U-Net, UNet++, TransUNet, Swin-UNet) using standardized protocols on two datasets with different anatomical compositions—a heterogeneous multi-region dataset (Ver1) and a mediastinal-focused dataset (Ver2). CST achieved the highest segmentation performance on anatomically focused data with Dice score 0.686 (Ver2), particularly excelling at small lymph nodes (<10 mm) with Dice 0.711 ± 0.149 , outperforming baseline approaches by 3.6–9.5 percentage points. However, on heterogeneous multi-region data (Ver1), UNet++ achieved the best performance (0.606), followed by U-Net (0.598) and Swin-UNet (0.582), demonstrating that architectural advantages depend on dataset composition. Size-stratified analysis revealed clinically acceptable results for larger lymph nodes (≥ 15 mm) reaching Dice coefficients up to 0.700, while performance dropped markedly for small nodes. Models trained and tested on single-region data (Ver2, mediastinal-only) achieved 5.7–13.6 percentage points higher Dice scores compared to models trained and tested on multi-region data (Ver1), demonstrating that anatomically focused segmentation tasks yield substantially higher performance than multi-region segmentation tasks across all evaluated architectures. These results provide evidence-based guidance for clinical implementation and establish baseline references for future lymph node segmentation research.

INDEX TERMS Cross-Slice Transformer, lymph node segmentation, CT imaging, adaptive slice fusion, 2.5D processing, deep learning, transformer attention

I. INTRODUCTION

LYMPH node metastasis remains one of the strongest prognostic indicators in oncology and has a direct impact on staging, treatment planning, and overall survival across almost every solid malignancy. In the widely used TNM staging system, nodal (N) status is given equal weight to

the primary tumor (T) and distant metastases (M)—this alone illustrates how crucial accurate nodal assessment is in daily practice. The presence, number, and location of metastatic nodes can completely change the therapeutic approach and provide critical clues about a patient's likely outcome [1].

In breast cancer, for example, the degree of axillary

involvement often decides whether a patient needs adjuvant chemotherapy, and how aggressive that treatment should be [2]. In lung cancer, finding metastatic mediastinal nodes typically rules out surgery and moves patients toward chemoradiation instead [3]. In gastroesophageal cancers, nodal burden and distribution guide the use of neoadjuvant therapy and the extent of resection [4]. Because these decisions have major effects on morbidity, quality of life, and healthcare costs, accurate lymph node assessment remains essential in modern oncology.

A. LIMITATIONS OF CURRENT CLINICAL PRACTICE

Computed tomography (CT) remains the primary imaging modality of choice for lymph node evaluation in routine oncologic practice because of its widespread availability, rapid acquisition, whole-body coverage, and superior spatial resolution. Current radiological assessment relies primarily on size criteria. Nodes larger than 10 mm in short-axis diameter are considered suspicious according to RECIST guidelines [5]. This size-based approach, however, has well-recognized limitations, with reported sensitivities of 57–76% and specificities of 55–82% across cancer types. The problem is bidirectional—micrometastases frequently occur in normal-sized nodes (false negatives), while reactive or inflammatory enlargement commonly causes benign nodes to exceed the threshold (false positives). In addition, manual identification and measurement across hundreds of slices is time-consuming and shows considerable inter-observer variability (15–35%) even among experienced radiologists [6].

B. TECHNICAL CHALLENGES IN AUTOMATED ANALYSIS

Automated lymph node analysis faces several distinct technical difficulties that set it apart from many other medical segmentation tasks. Lymph nodes are small (typically 3–30 mm), placing them near the resolution limit of clinical CT scanners. They show low contrast against surrounding tissues, with Hounsfield units similar to muscle, vessels, and other structures. They occupy less than 1% of voxels in whole-body scans, creating severe class imbalance that biases standard algorithms toward the background. There is also substantial inter-patient variability in node size, shape, texture, and anatomical distribution. Finally, lymph nodes often resemble adjacent vessels and muscle bundles, leading to frequent segmentation uncertainty even for expert readers.

C. DEEP LEARNING REVOLUTION IN MEDICAL IMAGING

Over the past decade, deep learning has driven major progress in medical image analysis. U-Net [7] established the now-standard encoder–decoder architecture with skip connections, enabling precise spatial localization while preserving multi-scale contextual information. ResNet [8] further demonstrated that residual connections facilitate the training of very deep networks by mitigating vanishing-gradient issues. Building on the success of U-Net, Zhou et al. [9] introduced UNet++, which redesigns the skip pathways between encoder and decoder using nested dense connections. This

architecture allows feature aggregation at multiple semantic levels and has proven particularly effective for medical image segmentation tasks involving large variability in target size and appearance.

More recently, transformer-based architectures have been successfully adapted for medical imaging. TransUNet [10] integrates a convolutional encoder with transformer blocks to combine fine-grained local features with long-range spatial dependencies. Swin Transformer [11] introduces a hierarchical representation with shifted-window self-attention, enabling efficient modeling of global context. Building on this design, Cao et al. [12] proposed Swin-UNet, which adopts a hierarchical Swin Transformer backbone for both encoding and decoding stages, demonstrating that transformer-dominant architectures can achieve competitive segmentation performance when appropriately tailored to medical imaging tasks.

Deep learning has also shown success in other cancer imaging domains. Recent work in mammography demonstrates that attention mechanisms, multi-scale feature extraction, and specialized loss functions can effectively address challenges related to small target sizes, low tissue contrast, and severe class imbalance [13]. These findings suggest that similar architectural principles may be beneficial for CT-based lymph node segmentation, despite inherent modality-specific differences.

When dealing with volumetric CT data, full 3D convolutional networks capture rich spatial context but quickly become computationally prohibitive due to high memory demands. A common alternative is the 2.5D strategy, which processes several adjacent slices simultaneously, striking a balance between volumetric awareness and computational efficiency [14], [15]. Cross-slice attention mechanisms provide an additional means of exchanging contextual information between neighboring slices without the overhead of full 3D processing [16]. Despite these advances, automated lymph node segmentation remains challenging due to extreme class imbalance, limited annotated data, high anatomical variability, and stringent clinical accuracy requirements.

D. CONTRIBUTIONS AND PAPER ORGANIZATION

Although deep learning has been widely applied to lymph node segmentation, most existing studies focus on individual architectures or limited dataset configurations, leaving several important aspects underexplored. In particular, the influence of anatomical dataset composition on segmentation performance across different architectural paradigms has not been systematically analyzed. Moreover, explicit inter-slice modeling using attention-based mechanisms within memory-efficient 2.5D frameworks remains relatively uncommon.

This work addresses these gaps through the following main contributions:

- 1) **Cross-Slice Transformer (CST):** We propose a Cross-Slice Transformer that automatically learns slice importance by combining statistical descriptors with transformer attention, stabilized through consistency reg-

ularization that aligns attention weights with natural inter-slice variations—a lightweight approach to incorporating volumetric context in 2.5D pipelines. The CST module is integrated with a U-Net decoder for end-to-end segmentation training.

- 2) **Dual-dataset evaluation:** We evaluate CST and four baseline segmentation architectures on two dataset configurations with contrasting anatomical coverage. Ver1 is a multi-region dataset, while Ver2 is a mediastinal-focused subset. This setup allows us to assess how anatomical homogeneity influences model performance and provides insights for clinical dataset curation.
- 3) **Systematic architectural comparison:** We benchmark CST against four state-of-the-art segmentation models representing multiple architectural paradigms: standard CNN (U-Net), nested CNN (UNet++), hybrid CNN-transformer (TransUNet), and pure transformer (Swin-UNet), establishing comprehensive baseline references across architectural families.
- 4) **Memory-efficient 2.5D pipeline:** All experiments employ a 2.5D strategy processing three consecutive CT slices, balancing inter-slice context with computational efficiency for training on standard hardware.

The rest of the paper is organized as follows. Section II reviews previous deep-learning work on lymph node segmentation. Section III describes the datasets, preprocessing, architectures, and training protocols. Section IV presents the experimental results. Section V discusses the findings, limitations, and potential clinical translation. Section VI concludes and outlines future research directions.

II. RELATED WORK

A. U-NET BASED ARCHITECTURES

U-Net and its variants remain the most widely used backbone for lymph node segmentation. Nayan et al. [17] introduced a modified UNet++ with total generalized variation upsampling and reported 94.8% accuracy, 91.9% Jaccard index, 94.1% recall, and 93.1% precision on mediastinal nodes from public datasets. The nested dense skip connections of UNet++ [9] are particularly helpful for handling the large size variation typical of lymph nodes.

Attention mechanisms have also become common. Manjunatha et al. [18] showed that modified U-Net with residual learning improved lymph node detection in CT scans by effectively handling small nodes in both abdominal and mediastinal regions. Al Hasan et al. [19] proposed a two-stage deep learning framework for automated segmentation of lymph nodes on neck CT scans, achieving high sensitivity while reducing false positives through 3D CNN refinement.

B. TRANSFORMER-BASED ARCHITECTURES

Hybrid CNN–transformer models have gained increasing attention in medical image segmentation. TransUNet [10] employs a ResNet-50 backbone for feature extraction, followed

by transformer layers embedded within a U-Net-style decoder, achieving 77.48% average Dice on the Synapse multi-organ dataset. Swin Transformer [11] introduced shifted-window self-attention and hierarchical feature representations to improve computational efficiency while capturing long-range dependencies. Building on this foundation, Swin-UNet [12] applies a transformer-dominant architecture to both encoder and decoder stages, demonstrating strong segmentation performance across medical imaging benchmarks.

Extending transformers to volumetric data, nnFormer [20] processes non-overlapping 3D patches using interleaved local and global attention, achieving a mean Dice score of 86.57% on multi-organ segmentation tasks. For lymph node-specific applications, Yu et al. [21] addressed positional bias in transformer-based models by introducing location-debiased query selection and contrastive query representations, resulting in improved lymph node segmentation performance in CT scans. Park et al. [22] employed DenseNet121 [23] as a feature extractor and constructed an ensemble classifier to predict axillary lymph node metastasis directly from primary breast tumor images, achieving an AUROC of 0.968 and sensitivity of 0.980.

C. 2.5D PROCESSING STRATEGIES

Because full 3D convolutions quickly exhaust GPU memory on clinical CT volumes, most practical works rely on 2.5D approaches that feed several neighboring slices into the network at once. For instance, Kumar et al. [14] developed a flexible 2.5D medical image segmentation approach with in-slice and cross-slice attention, achieving high performance on volumetric segmentation tasks. Roth et al. [15] trained deep CNNs on randomly oriented 2.5D patches for lymph node detection, and Wang et al. [16] enriched ordinary 2D networks by learning explicit contextual embeddings from adjacent slices. These approaches demonstrate that effective inter-slice modeling can be achieved without the computational burden of full 3D processing.

D. LOSS FUNCTIONS FOR CLASS IMBALANCE

Extreme foreground-background imbalance is a persistent issue in lymph node segmentation. Focal Loss [24] focuses training on hard examples, Tversky Loss [25] directly balances false positives and false negatives, and Focal Tversky Loss [26] merges both ideas. Yeung et al. [27] later proposed Unified Focal Loss as a more flexible framework. In practice, most recent lymph node segmentation studies combine Focal and Dice losses to achieve stable convergence and address the severe class imbalance inherent in medical imaging tasks.

III. METHODOLOGY

A. OVERVIEW OF EXPERIMENTAL DESIGN

Fig. 1 presents our experimental pipeline. The workflow encompasses four main stages: (1) preprocessing with body-centered cropping, intensity windowing, and 2.5D slice extraction; (2) dataset configurations using two anatomical compositions with patient-level train/validation/test splitting;

(3) model training using our proposed CST alongside four baseline segmentation architectures (U-Net, UNet++, TransUNet, Swin-UNet); and (4) evaluation producing pixel-level segmentation masks with comprehensive performance metrics.

B. DATASET CHARACTERISTICS

This study employed two datasets with contrasting anatomical compositions. Ver1 [28] contains 21,906 annotated CT slices from 689 patients across multiple institutions, covering mediastinal, abdominal, and pelvic lymph nodes. Ver2 [29] contains 15,056 slices from 603 patients, restricted to mediastinal lymph nodes only. Both datasets used patient-level splitting (70%/15%/15% for train/validation/test) to prevent data leakage.

Ver1 splits: 482 training patients (14,876 slices), 103 validation (3,196 slices), 104 test (3,834 slices). Ver2 splits: 422 training patients (10,699 slices), 90 validation (2,052 slices), 91 test (2,305 slices).

All CT scans were acquired using standard clinical protocols with slice thickness 1.0–5.0 mm and image dimensions 512×512 to 768×768 pixels. Ground truth annotations were produced by experienced radiologists.

C. PREPROCESSING PIPELINE

All CT images and segmentation masks underwent a standardized preprocessing pipeline to ensure consistency across heterogeneous datasets. The complete preprocessing workflow is illustrated in Fig. 2 and comprises four sequential stages.

1) Body-Centered Region of Interest Extraction

Raw CT volumes contain substantial non-anatomical background regions that provide no diagnostic information. We implemented an automated body center detection algorithm operating on CT Hounsfield Unit (HU) values. For each axial slice, a binary body mask was generated by applying an intensity threshold at -200 HU, which effectively separates tissue from air. The geometric center of the body mask was computed as the mean of all body pixel coordinates. A 384×384 pixel region was extracted around this center to focus computational resources on anatomically relevant areas while maintaining sufficient context.

2) Intensity Windowing and Normalization

To enhance lymph node visibility and improve contrast, we applied anatomically appropriate windowing based on clinical radiology standards. For mediastinal imaging, we employed window level (WL) = 40 HU and window width (WW) = 400 HU, optimized for soft tissue visualization. For abdominal imaging, WL = 50 HU and WW = 350 HU were used. Intensities outside the window range were clipped, and the windowed values were linearly rescaled to the range [0, 255].

3) Memory-Efficient 2.5D Representation

We adopted a memory-efficient 2.5D representation strategy that balances volumetric context with computational efficiency and storage requirements. For each target slice at index i , we extract three consecutive axial slices at positions $(i - 1)$, (i) , and $(i + 1)$, forming the input tensor:

$$\mathbf{X} = [I_{i-1}, I_i, I_{i+1}] \in \mathbb{R}^{3 \times H \times W} \quad (1)$$

where H and W represent height and width dimensions. For boundary slices where $(i - 1)$ or $(i + 1)$ are unavailable, the target slice is replicated to maintain consistent input dimensions.

Images were stored in a compressed lossless format to reduce storage requirements while preserving diagnostic quality. Combined with 2.5D processing of only three slices, our approach maintains inter-slice context while remaining computationally tractable on standard GPUs.

4) Spatial Resizing and Channel Stacking

Image patches of 384×384 pixels were resized to 256×256 pixels to balance spatial resolution with computational efficiency. Bilinear interpolation was used for CT images to preserve intensity gradients, while nearest-neighbor interpolation was applied to segmentation masks to maintain label integrity. The three consecutive slices were stacked along the channel dimension to form the final 256×256 input, where $\mathbf{Y}_i \in \{0, 1\}^{256 \times 256}$ represents the binary segmentation mask for the center slice. Images were normalized using ImageNet statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]) to leverage pretrained encoder weights.

D. MODEL ARCHITECTURES

We propose a Cross-Slice Transformer (CST) for adaptive inter-slice modeling in 2.5D CT analysis and compare its performance against four baseline segmentation architectures representing diverse design paradigms: standard CNN (U-Net), nested CNN (UNet++), hybrid CNN-transformer (TransUNet), and pure transformer (Swin-UNet).

1) U-Net

U-Net [7] with ResNet-50 [8] encoder pretrained on ImageNet provides the baseline CNN segmentation architecture. We employed the implementation from `segmentation_models_pytorch` [30]. The encoder comprises five stages with progressively increasing feature dimensionality (64, 256, 512, 1024, 2048 channels) that extract hierarchical representations. The decoder upsamples features through four stages using bilinear interpolation followed by convolutional layers. Skip connections between corresponding encoder-decoder levels preserve spatial details. Input size: 256×256 .

2) UNet++

UNet++ [9] extends U-Net through nested, dense skip connections between encoder and decoder. We used the `segmentation_models_pytorch` implementation

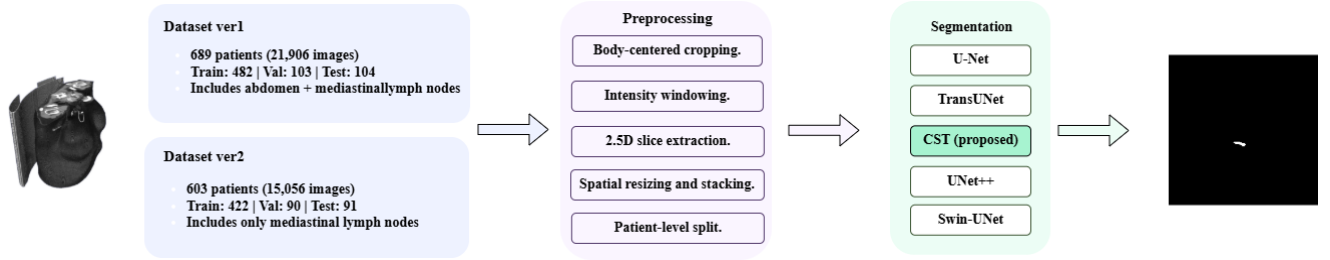


FIGURE 1. Experimental pipeline for Cross-Slice Transformer and baseline segmentation architectures. The workflow encompasses preprocessing with body-centered cropping and 2.5D slice extraction, model training across two anatomical configurations (multi-region Ver1 and mediastinal-focused Ver2), and comprehensive evaluation using patient-level metrics.

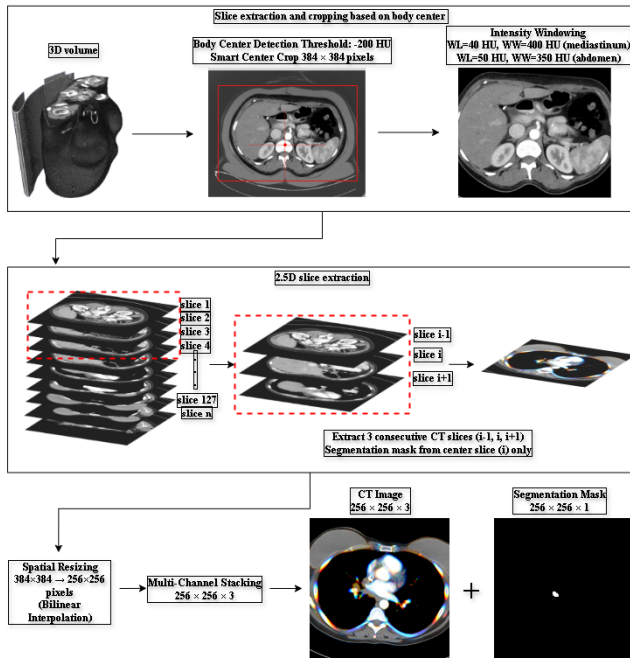


FIGURE 2. Comprehensive preprocessing pipeline for 2.5D lymph node segmentation showing body-centered ROI extraction, intensity windowing, multi-slice extraction, and spatial resizing.

with ResNet-34 encoder pretrained on ImageNet. Intermediate decoder nodes aggregate features from multiple encoder levels, enabling multi-scale feature fusion particularly beneficial for objects with substantial size variation. Input size: 256×256 .

3) TransUNet

TransUNet [10] combines CNN feature extraction with transformer-based global context modeling. We implemented a hybrid architecture using ResNet-50 CNN encoder from `torchvision` and Vision Transformer (ViT-Base-Patch16) from `timm` [31], both pretrained on ImageNet. The ViT backbone employs 12 transformer layers with 12 attention heads, 768-dimensional embeddings, and patch size 16. The CNN decoder with channel progression [256, 128, 64, 16] performs spatial upsampling for final segmentation. Input size: 224×224 .

4) Swin-UNet

Swin-UNet [12] integrates hierarchical Swin Transformer blocks with shifted-window attention as the encoder. We implemented a hybrid architecture using `segmentation_models_pytorch`, combining a Swin-Tiny encoder pretrained on ImageNet with a standard U-Net CNN decoder. The encoder uses window size 7, patch size 4, and [2, 2, 6, 2] transformer blocks per stage with [3, 6, 12, 24] attention heads and 96-dimensional embeddings at each stage. The decoder employs bilinear upsampling with convolutional blocks and channel progression [256, 128, 64, 32, 16] for spatial reconstruction. Input size: 224×224 .

5) Cross-Slice Transformer (CST)

The Cross-Slice Transformer (CST) learns adaptive slice importance through statistical descriptor extraction and transformer-based attention for explicit inter-slice modeling. The architecture consists of three main components integrated with a U-Net decoder for end-to-end segmentation, as illustrated in Figure 3.

Statistical Descriptor Extraction: For each input comprising three slices $\mathbf{X} \in \mathbb{R}^{B \times 3 \times H \times W}$, we extract compact statistical descriptors that encode slice-level intensity distribution and inter-slice coherence. For each slice $s \in \{1, 2, 3\}$, we compute mean intensity μ_s , standard deviation σ_s , and inter-slice difference δ_s :

$$\begin{aligned} \mu_s &= \frac{1}{HW} \sum_{h,w} X_{s,h,w} \\ \sigma_s &= \sqrt{\frac{1}{HW} \sum_{h,w} (X_{s,h,w} - \mu_s)^2} \\ \delta_s &= \begin{cases} \frac{1}{HW} \sum_{h,w} |X_{2,h,w} - X_{1,h,w}| & \text{if } s = 1 \\ \frac{1}{HW} \sum_{h,w} |X_{s,h,w} - X_{s-1,h,w}| & \text{if } s > 1 \end{cases} \end{aligned} \quad (2)$$

Transformer-Based Attention: The descriptor vectors are embedded into a higher-dimensional space through a linear layer followed by LayerNorm and GELU activation, then processed through a transformer encoder:

$$\begin{aligned} \mathbf{T}_s &= \text{GELU}(\text{LN}(\mathbf{W}_e \mathbf{d}_s)) \\ \mathbf{T}' &= \text{TransformerEncoder}([\mathbf{T}_1, \mathbf{T}_2, \mathbf{T}_3]) \\ w_s &= \frac{\exp(\mathbf{W}_w \mathbf{T}'_s)}{\sum_{j=1}^3 \exp(\mathbf{W}_w \mathbf{T}'_j)} \end{aligned} \quad (3)$$

where $\mathbf{W}_e \in \mathbb{R}^{96 \times 4}$ is the embedding matrix, $\mathbf{W}_w \in \mathbb{R}^{1 \times 96}$ generates attention logits, and w_s represents the learned importance weight for slice s . The transformer encoder employs 2 layers with 4 attention heads, hidden dimension 96, feed-forward dimension 192, and dropout 0.1.

Contextual Fusion and Feature Extraction: The learned weights are applied to create an adaptively weighted slice fusion, combined with spatial difference maps:

$$\begin{aligned} \mathbf{F}_{fused} &= \sum_{s=1}^3 w_s \cdot \mathbf{X}_s \\ \mathbf{D}_{spatial} &= \frac{1}{2} \sum_{s=2}^3 |\mathbf{X}_s - \mathbf{X}_{s-1}| \\ \mathbf{C} &= [\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3, \mathbf{F}_{fused}, \mathbf{D}_{spatial}] \in \mathbb{R}^{B \times 5 \times H \times W} \end{aligned} \quad (4)$$

The concatenated context $\mathbf{C}$ is processed through convolutional layers to extract features that are then fed to a U-Net decoder with 32 base channels and channel progression [32, 64, 128, 256] across 3 downsampling levels. Input size: 256×256 .

Consistency Regularization: To encourage the learned attention weights to align with inter-slice variation patterns, we introduce a consistency regularization term. We compute normalized difference energy:

$$e_s = \frac{\delta_s}{\sum_{j=1}^3 \delta_j + \epsilon} \quad (5)$$

where $\epsilon = 10^{-6}$ prevents division by zero. The regularization loss enforces agreement between learned weights and difference energy:

$$\mathcal{L}_{consistency} = \frac{1}{3} \sum_{s=1}^3 (w_s - e_s)^2 \quad (6)$$

This regularization encourages the model to assign higher weights to slices with greater information content relative to adjacent slices.

Table 1 presents the computational complexity of all evaluated architectures. CST achieves the most compact design with only 2.10M parameters—the next smallest model (UNet++) has $12\times$ more parameters—while maintaining competitive FLOPs (10.54G). This efficiency stems from the lightweight statistical descriptor extraction and compact transformer attention mechanism, making CST particularly suitable for clinical deployment with limited computational resources. Despite its simplicity, CST demonstrates that explicit inter-slice modeling through adaptive attention can be achieved without the parameter overhead of large pretrained transformers (TransUNet: 99.40M parameters) or hierarchical architectures (Swin-UNet: 31.63M parameters).

E. TRAINING CONFIGURATION

To ensure rigorous and fair comparison, we employed a controlled experimental design where all models used identical training configurations across both datasets. This approach

TABLE 1. Computational complexity comparison of all segmentation architectures evaluated in this study.

Model	Parameters	FLOPs	Efficiency Ratio
CST (Ours)	2.10M	10.54G	1.00×
Swin-UNet	31.63M	6.76G	15.06× params
TransUNet	99.40M	24.68G	47.33× params
U-Net	32.52M	10.71G	15.49× params
UNet++	26.08M	18.45G	12.42× params

isolates the impact of architectural differences and anatomical dataset composition from confounding effects of model-specific hyperparameter tuning, enabling direct attribution of performance differences to the core contributions of each architecture.

1) Unified Training Protocol

All segmentation models (CST, U-Net, UNet++, TransUNet, Swin-UNet) were trained with identical hyperparameters on both Ver1 and Ver2 datasets: AdamW optimizer with learning rate 1×10^{-4} , weight decay 1×10^{-4} , batch size 8, and 100 training epochs. This unified protocol ensures that observed performance differences reflect inherent architectural capabilities rather than optimization artifacts.

2) Loss Function Design

To address the extreme class imbalance inherent in lymph node segmentation (lymph nodes occupy less than 1% of image volume), all segmentation models employed a combined loss function comprising Focal Loss and Dice Loss with equal weighting:

$$\mathcal{L}_{seg} = 0.5 \cdot \mathcal{L}_{Focal} + 0.5 \cdot \mathcal{L}_{Dice} \quad (7)$$

Focal Loss addresses class imbalance by down-weighting easy examples and focusing training on hard, misclassified cases:

$$\begin{aligned} \mathcal{L}_{Focal} = -\frac{1}{N} \sum_{i=1}^N & \left[\alpha(1-p_i)^\gamma y_i \log(p_i) \right. \\ & \left. + (1-\alpha)p_i^\gamma (1-y_i) \log(1-p_i) \right] \end{aligned} \quad (8)$$

where p_i is the predicted probability for pixel i , $y_i \in \{0, 1\}$ is the ground truth label, $\alpha = 0.75$ balances positive and negative classes, and $\gamma = 2.0$ controls the rate at which easy examples are down-weighted. The focusing parameter γ reduces the loss contribution from well-classified examples, enabling the model to concentrate on challenging cases such as small lymph nodes with ambiguous boundaries.

Dice Loss directly optimizes the overlap between predicted and ground truth masks, providing complementary robustness to class imbalance:

$$\mathcal{L}_{Dice} = 1 - \frac{2 \sum_{i=1}^N p_i y_i + \epsilon}{\sum_{i=1}^N p_i + \sum_{i=1}^N y_i + \epsilon} \quad (9)$$

where $\epsilon = 1.0$ is a smoothing factor that prevents numerical instability when both prediction and ground truth are empty.

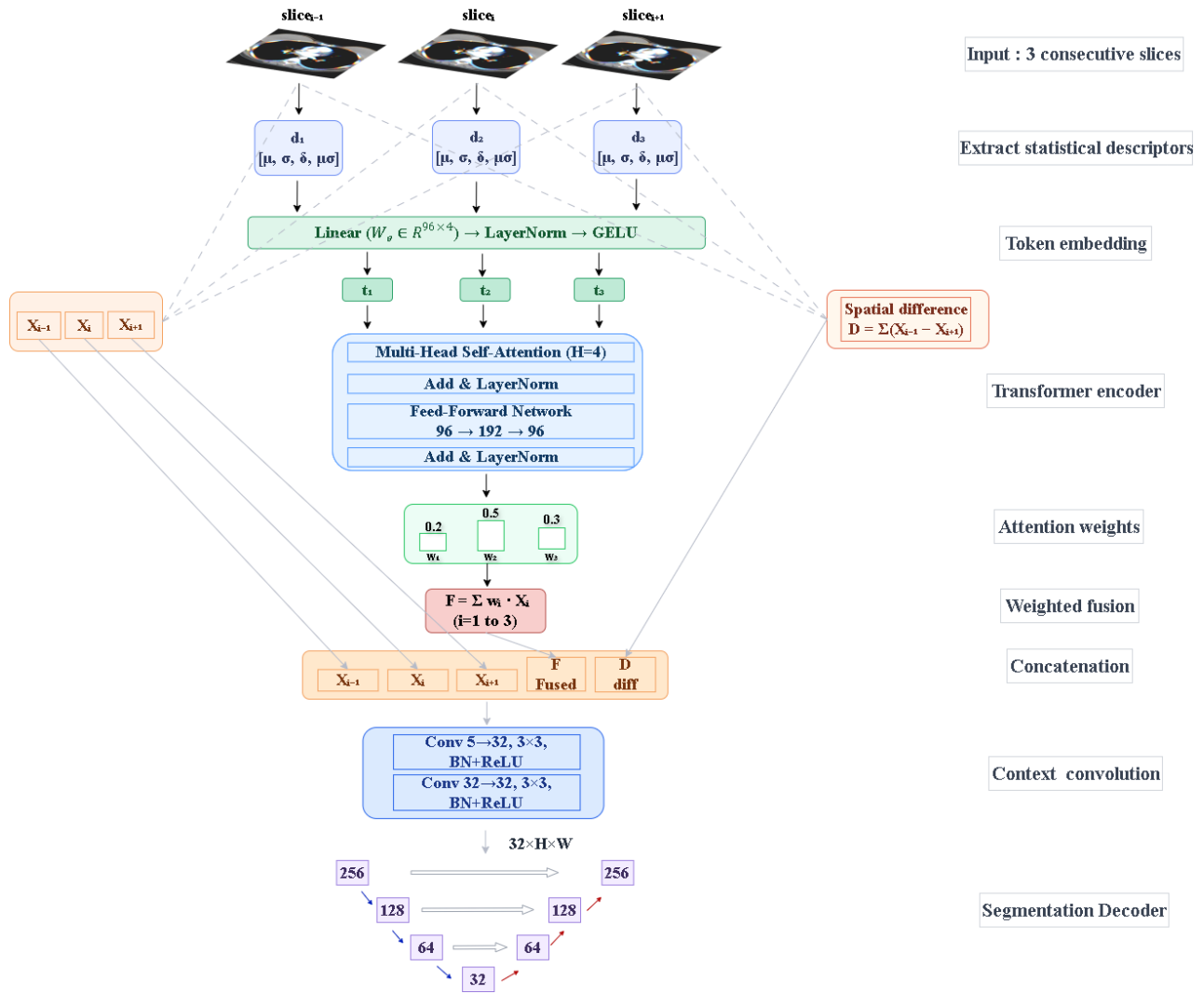


FIGURE 3. Architecture of the Cross-Slice Transformer (CST). The model extracts statistical descriptors from three consecutive CT slices, computes slice importance weights via transformer attention, and fuses slices adaptively before U-Net decoding for final segmentation.

Unlike pixel-wise losses, Dice Loss evaluates region-level overlap, making it particularly effective for segmentation tasks with severe foreground-background imbalance.

For CST, we additionally incorporated a consistency regularization term that aligns attention weights with inter-slice variation patterns:

$$\mathcal{L}_{CST} = 0.5 \cdot \mathcal{L}_{Focal} + 0.5 \cdot \mathcal{L}_{Dice} + 0.1 \cdot \mathcal{L}_{consistency} \quad (10)$$

where $\mathcal{L}_{consistency}$ is defined in Equation 6.

The combination of Focal Loss and Dice Loss provides synergistic benefits: Focal Loss ensures precise pixel-level classification while handling class imbalance through adaptive example weighting, whereas Dice Loss enforces global region overlap and maintains segmentation coherence. The equal weighting (0.5 each) balances both objectives without requiring task-specific tuning. This formulation has proven effective for medical segmentation tasks involving small, rare structures with challenging boundaries [26], [27].

3) Optimization and Regularization

All models were trained using AdamW optimizer [32] with $\beta_1 = 0.9$, $\beta_2 = 0.999$, learning rate 1×10^{-4} , and weight decay 1×10^{-4} for regularization. Learning rate scheduling employed ReduceLROnPlateau with factor 0.5 and patience 5 epochs, reducing the learning rate when validation Dice score plateaued. Early stopping with patience 15 epochs monitored validation Dice score to prevent overfitting.

Data augmentation was applied identically across all models to enhance robustness while preserving anatomical validity. Augmentations included geometric transformations (random rotation $\pm 10^\circ$, shift-scale transformation with limits 0.05, elastic deformations with $\sigma=8$ and $\sigma=4$) and intensity transformations (random brightness and contrast adjustment ± 0.1), each applied with probability 0.3-0.5.

All models were trained once with these fixed hyperparameters and evaluated on held-out test sets to ensure unbiased performance estimates. This single-run protocol with-

out model-specific hyperparameter tuning reflects realistic clinical deployment scenarios and enables fair architectural comparison by isolating performance differences to model design rather than optimization choices.

F. EVALUATION METRICS

1) Segmentation Metrics

Dice coefficient served as the primary evaluation metric for segmentation models, measuring overlap between predicted and ground truth masks ranging from 0 (no overlap) to 1 (perfect agreement):

$$\text{Dice} = \frac{2|P \cap G|}{|P| + |G|} \quad (11)$$

where P represents the predicted segmentation and G the ground truth.

Patient-Level Metric Aggregation: All segmentation metrics were computed at the patient level to provide clinically relevant performance estimates. For each patient, metrics were first computed for each slice containing lymph nodes, then averaged across all slices within that patient to obtain a patient-level score. Final reported metrics represent the mean and standard deviation across all test patients. This patient-level aggregation ensures equal contribution from each patient regardless of the number of slices they contribute, avoiding bias toward patients with more extensive disease or larger scan ranges.

Mathematically, for N test patients where patient i has S_i annotated slices:

$$\text{Dice}_{\text{reported}} = \frac{1}{N} \sum_{i=1}^N \left(\frac{1}{S_i} \sum_{j=1}^{S_i} \text{Dice}_{i,j} \right) \quad (12)$$

Additional metrics included Intersection over Union (IoU), precision (positive predictive value), and recall (sensitivity) to provide complementary perspectives on segmentation quality:

$$\text{IoU} = \frac{|P \cap G|}{|P \cup G|} \quad (13)$$

$$\text{Precision} = \frac{|P \cap G|}{|P|} \quad (14)$$

$$\text{Recall} = \frac{|P \cap G|}{|G|} \quad (15)$$

All metrics were aggregated using the same patient-level approach described above.

Size-Stratified Analysis: To assess model performance across clinically relevant size categories, we stratified results by lymph node size: <10 mm, 10–15 mm, and 15 mm. Size was determined from ground truth annotations by computing the short-axis diameter in the axial plane using the minimum dimension of the minimum area bounding rectangle. This stratification reveals model capabilities for detecting small versus large lymph nodes, providing clinically actionable insights for deployment.

IV. RESULTS

A. OVERALL MODEL PERFORMANCE

Table 3 presents overall segmentation performance across all test set lymph nodes. On anatomically focused Ver2 data, CST achieved the highest performance (Dice 0.686). However, on heterogeneous Ver1 data, UNet++ achieved the best performance (0.606), followed by U-Net (0.598), Swin-UNet (0.582), and CST (0.569). All models showed substantial Ver1-to-Ver2 improvements: TransUNet (+13.6 points), CST (+11.7), Swin-UNet (+6.6), U-Net (+6.4), and UNet++ (+5.7).

This consistent improvement pattern across all architectures (5.7–13.6 percentage points) when training and testing on mediastinal-only data (Ver2) compared to multi-region data (Ver1) demonstrates that single-anatomy segmentation tasks are substantially less challenging than multi-region tasks, regardless of model architecture. The mediastinal-only configuration reduces anatomical variability, simplifies tissue context variations, and provides more consistent training signals. Transformer-based architectures (TransUNet, CST) showed the largest improvements (11.7–13.6 points), while CNN-based architectures showed smaller gains (5.7–6.6 points). This suggests that attention mechanisms may be particularly effective at exploiting reduced anatomical complexity, though all architectures benefit substantially from the simplified single-region task.

CST on Ver2 achieved the highest recall (0.765) combined with excellent precision (0.680), contributing to its best-in-class Dice coefficient. UNet++ demonstrated strong balanced performance (0.658 precision, 0.661 recall), while TransUNet and Swin-UNet showed complementary precision-recall trade-offs (see Table 3).

The performance levels ranging from Dice 0.485 to 0.686 reflect substantial progress through architectural innovations and anatomically focused training, while highlighting remaining challenges. Despite CST's best-in-class performance, multiple factors continue to limit absolute performance: (1) extreme class imbalance (<1% foreground voxels), (2) low contrast between lymph nodes and surrounding soft tissues, (3) high anatomical variability, (4) morphological similarity to vessels and muscle, and (5) limited training data. These challenges manifest across all architectures, indicating that substantial further improvements require larger multi-institutional datasets, adaptive loss weighting strategies, and potentially multi-modal imaging integration.

B. SIZE-STRATIFIED ANALYSIS

Table 2 presents segmentation performance stratified by lymph node size, revealing that all models face substantial challenges with small nodes while achieving clinically acceptable performance on larger nodes.

1) Small Lymph Nodes (<10mm)

Small lymph nodes (<10 mm) present the most significant challenge across all architectures. On Ver1, UNet++ achieved the highest Dice (0.631 ± 0.205), followed by U-Net ($0.618 \pm$

TABLE 2. Size-stratified segmentation performance on test sets (patient-level). Lymph nodes categorized by short-axis diameter. Values shown as mean $\pm$ standard deviation. Best Dice and IoU scores per size category and dataset shown in bold.

Model	Dataset	Size Range	Dice	IoU
CST	Ver1	<10 mm	0.589 ± 0.193	0.504 ± 0.176
		10–15 mm	0.509 ± 0.177	0.416 ± 0.152
		≥ 15 mm	0.588 ± 0.163	0.481 ± 0.147
	Ver2	<10 mm	0.711 ± 0.149	0.620 ± 0.147
		10–15 mm	0.617 ± 0.166	0.517 ± 0.157
		≥ 15 mm	0.680 ± 0.165	0.577 ± 0.172
U-Net	Ver1	<10 mm	0.618 ± 0.219	0.540 ± 0.204
		10–15 mm	0.538 ± 0.209	0.454 ± 0.190
		≥ 15 mm	0.618 ± 0.146	0.515 ± 0.133
	Ver2	<10 mm	0.675 ± 0.146	0.587 ± 0.146
		10–15 mm	0.621 ± 0.172	0.524 ± 0.160
		≥ 15 mm	0.676 ± 0.136	0.573 ± 0.149
UNet++	Ver1	<10 mm	0.631 ± 0.205	0.551 ± 0.195
		10–15 mm	0.543 ± 0.238	0.463 ± 0.220
		≥ 15 mm	0.606 ± 0.172	0.506 ± 0.158
	Ver2	<10 mm	0.671 ± 0.145	0.585 ± 0.144
		10–15 mm	0.622 ± 0.162	0.530 ± 0.152
		≥ 15 mm	0.700 ± 0.121	0.591 ± 0.137
TransUNet	Ver1	<10 mm	0.466 ± 0.184	0.385 ± 0.166
		10–15 mm	0.506 ± 0.188	0.420 ± 0.170
		≥ 15 mm	0.553 ± 0.157	0.449 ± 0.144
	Ver2	<10 mm	0.616 ± 0.164	0.529 ± 0.161
		10–15 mm	0.616 ± 0.149	0.519 ± 0.144
		≥ 15 mm	0.663 ± 0.120	0.554 ± 0.134
Swin-UNet	Ver1	<10 mm	0.590 ± 0.205	0.506 ± 0.191
		10–15 mm	0.551 ± 0.203	0.463 ± 0.187
		≥ 15 mm	0.613 ± 0.136	0.506 ± 0.125
	Ver2	<10 mm	0.655 ± 0.145	0.564 ± 0.142
		10–15 mm	0.619 ± 0.178	0.525 ± 0.166
		≥ 15 mm	0.668 ± 0.140	0.565 ± 0.149

0.219). Ver2 training dramatically improved performance, with CST reaching Dice 0.711 ± 0.149 (+12.2 points), outper-

forming all baselines by 4.0–9.5 percentage points, highlighting the combined benefits of cross-slice attention and anatom-

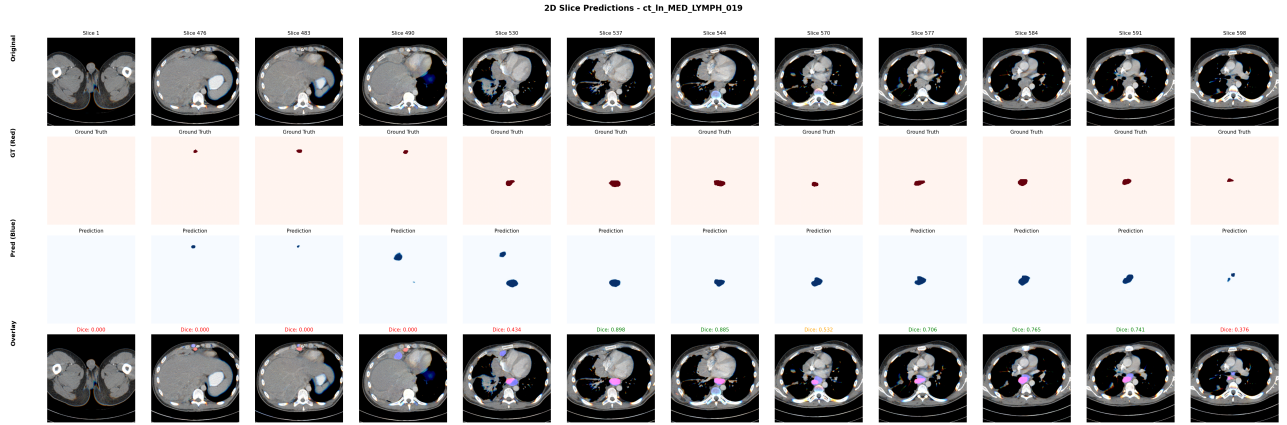


FIGURE 4. Qualitative comparison of segmentation predictions. CST demonstrates superior performance on small lymph nodes (<10 mm) while all models achieve acceptable results on larger nodes (≥ 15 mm). Ground truth annotations shown in green, model predictions in red.

TABLE 3. Overall segmentation performance on test sets (patient-level). Metrics computed across all lymph nodes regardless of size. Best Dice score per dataset shown in bold.

Model	Dataset	Dice	IoU	Precision	Recall
CST	Ver1	0.5690	0.4793	0.5421	0.6861
CST	Ver2	0.6857	0.5909	0.6803	0.7647
U-Net	Ver1	0.5983	0.5162	0.6020	0.6548
U-Net	Ver2	0.6625	0.5706	0.6540	0.7429
UNet++	Ver1	0.6063	0.5242	0.6167	0.6577
UNet++	Ver2	0.6631	0.5733	0.6737	0.7240
TransUNet	Ver1	0.4852	0.4005	0.4993	0.5245
TransUNet	Ver2	0.6217	0.5300	0.6212	0.6919
Swin-UNet	Ver1	0.5823	0.4949	0.5871	0.6318
Swin-UNet	Ver2	0.6483	0.5554	0.6341	0.7370

ical homogeneity for detecting challenging, subtle structures.

2) Medium-Sized Lymph Nodes (10–15mm)

Medium-sized lymph nodes (10–15 mm) demonstrate improved segmentation across all models. Ver1 results ranged from Dice 0.506 to 0.551, indicating reasonable performance even on heterogeneous data. Ver2 training yielded substantial improvements, with all models converging between 0.616–0.622, demonstrating minimal architectural differences for this size category. UNet++ achieved the highest Dice (0.622 ± 0.162), followed by U-Net (0.621 ± 0.172) and Swin-UNet (0.619 ± 0.178). These gains demonstrate that anatomically homogeneous training particularly enhances medium node detection across both CNN and transformer architectures.

3) Large Lymph Nodes (≥ 15 mm)

Large lymph nodes (≥ 15 mm) yield the best segmentation performance across all models, with Ver2 Dice scores converging between 0.663 and 0.700. UNet++ achieved the highest performance (0.700 ± 0.121), followed by CST (0.680 ± 0.165) and U-Net (0.676 ± 0.136). This convergence indicates that architectural differences have less impact for

well-defined, large structures, and that clinical requirements should guide model selection based on the target size distribution.

4) Clinical Implications

CST achieves the best overall Dice on Ver2 (0.686) because small lymph nodes (<10 mm) constitute the majority of test instances, and CST's substantial advantage in this category (Dice 0.711 ± 0.149) dominates patient-level averages. However, UNet++ demonstrates superior performance on medium (0.622 ± 0.162) and large (0.700 ± 0.121) nodes individually. On Ver1, UNet++ achieved the best overall performance (0.606), demonstrating robustness to anatomical heterogeneity. This pattern underscores the value of size-stratified analysis and dataset composition for understanding model capabilities beyond aggregate metrics.

Architecture selection should be guided by clinical requirements: CST for anatomically focused small node detection (dedicated mediastinal staging), UNet++ for robust multi-region general-purpose segmentation, and U-Net for computationally efficient deployment. The consistent 5.7–13.6 point improvements from Ver1 to Ver2 demonstrate that anatomical homogeneity benefits all architectures, with transformer-based models showing the largest gains.

V. DISCUSSION

A. KEY FINDINGS

CST achieved the highest overall Dice on Ver2 (0.686), demonstrating that cross-slice attention with consistency regularization effectively leverages inter-slice context in 2.5D CT analysis. However, on Ver1, UNet++ (0.606) and U-Net (0.598) outperformed CST (0.569), revealing that architectural advantages depend on dataset composition. Notably, CST's strong performance on small lymph nodes (<10 mm, Dice 0.711 ± 0.149 on Ver2) is the key reason it achieves the highest Ver2 Dice, as small nodes make up the majority of our test set, and this size category dominates the patient-level average. This small-node advantage clearly distinguishes CST

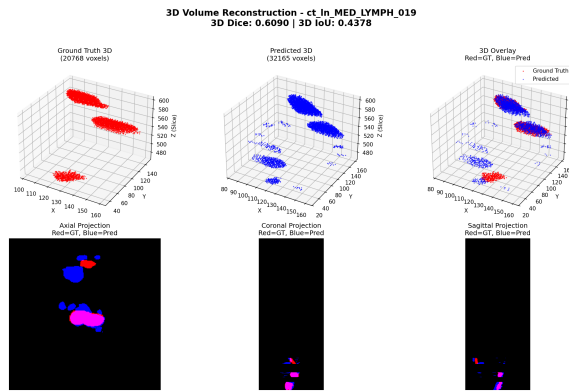


FIGURE 5. 3D volume construction during inference showing slice-by-slice prediction aggregation. The model processes consecutive axial slices using a 2.5D approach (three slices as input channels), generates predictions for each center slice, and aggregates them to form complete volumetric segmentation masks. This approach maintains inter-slice context while remaining computationally efficient compared to full 3D processing.

on anatomically homogeneous data, even when UNet++ performs better on medium and large nodes or on heterogeneous multi-region data.

B. IMPACT OF DATASET ANATOMICAL CONFIGURATION

Performance improvements from Ver2 over Ver1 (5.7–13.6 percentage points across all architectures) reflect that single-region segmentation tasks are less challenging than multi-region segmentation tasks. Ver2 models were trained and tested exclusively on mediastinal lymph nodes, while Ver1 models were trained and tested on multi-region data (mediastinal, abdominal, pelvic). The reduced anatomical variability in Ver2 simplifies the segmentation problem. Transformer-based models (TransUNet +13.6%, CST +11.7%) benefit most, while CNN-based models benefit less (UNet++ +5.7%, U-Net +6.4%). This suggests that region-specific, homogeneous training reduces confounding factors and improves feature learning, particularly for attention-based architectures.

C. CHALLENGES AND LIMITATIONS

Statistical Validation: Each model was trained once with fixed hyperparameters. Multiple training runs with different random seeds and cross-validation would provide more robust statistical validation with confidence intervals. The reported metrics represent single-run results rather than statistically validated population estimates.

2.5D Processing Limitations: Small lymph nodes (<1% of voxels) and inter-observer variability (15–35%) limit performance. The 2.5D approach may miss long-range context; full 3D models could improve results but require more computation. Ver2’s mediastinal-only focus also limits generalizability.

Clinical Validation: Our evaluation relies on comparison against reference annotations without radiologist review, inter-observer agreement analysis, or clinical reader studies.

Multi-center external validation and clinical evaluation are essential for deployment.

D. CLINICAL TRANSLATION

Deployment requires interpretability (attention/saliency maps), uncertainty quantification, PACS integration, and multi-modal fusion. Multi-center validation is essential for technical accuracy and patient benefit.

E. FUTURE DIRECTIONS

Future work should explore federated learning, self-supervised pretraining, longitudinal modeling, radiomics/genomics integration, active learning, adaptive loss weighting schemes and boundary-aware loss terms, and semi-supervised/ensemble methods to enhance small node detection and overall segmentation performance.

VI. CONCLUSION

We proposed the Cross-Slice Transformer (CST) for 2.5D lymph node segmentation. CST achieved the highest performance on anatomically focused Ver2 data (Dice 0.686), particularly excelling at small lymph nodes (Dice 0.711 ± 0.149). However, on heterogeneous Ver1 data, UNet++ achieved the best performance (0.606), followed by U-Net (0.598) and CST (0.569), demonstrating that architectural advantages depend on dataset composition. Single-region segmentation tasks (Ver2) yield 5.7–13.6 percentage points higher performance than multi-region tasks (Ver1) across all architectures, reflecting reduced anatomical complexity.

Size-stratified analysis guides model choice: CST for small nodes, U-Net for medium-to-large, and UNet++ for general-purpose use. Absolute performance on small nodes remains moderate, highlighting the need for larger datasets, enhanced loss function strategies, and clinical validation. While our controlled comparison provides valuable insights into architectural design principles, validation against nnU-Net, multi-center external testing, and clinical reader studies are essential next steps for assessing practical clinical utility and deployment readiness. Future work should continue to improve small node detection and segmentation reliability to support cancer staging and treatment planning.

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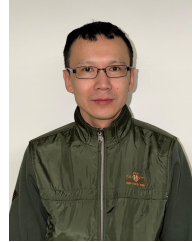
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